

Localization of Adenosine Receptor Messenger RNAs in the Rat Eye

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Adenosine is present in all cells and body fluids and has been suggested to play several roles in the physiology of ocular tissues. The present study was undertaken to determine which types of adenosine receptor mRNAs are present in the rat eye, and where they are expressed. RNA or deoxyoligonucleotides complementary to rat adenosine receptor subtypes A_1 , A_{2A} , A_{2B} and A_3 were used to generate ^{35}S labeled antisense and sense probes. The probes were then used for in situ hybridization on 10 μm cryosections of the rat eye including the cornea, iris, ciliary body, lens, retina, choroid and sclera. A_1 , A_{2A} and A_{2B} receptor mRNAs were demonstrated in the ciliary processes. A_1 receptor mRNA was also expressed in the ganglion cell layer of the retina. The retina also showed A_{2A} receptor mRNA expression, which was most prominent in the inner nuclear layer and less prominent in the ganglion cell layer and outer nuclear layer. Weak A_{2A} expression was found in the retinal pigment epithelium and choriocapillaris. No significant expression of A_3 receptor mRNA was found in the rat eye. In conclusion, using in situ hybridization, we have demonstrated expression of mRNA for A_1 , A_{2A} and A_{2B} adenosine receptors in the rat eye. The expression patterns support specific roles for adenosine in the ciliary process and retina.

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Key words: adenosine receptors; messenger RNA; in situ hybridization; retina; ciliary process.

1. Introduction

Adenosine is present in all cells and body fluids and is known to function as a modulator of a variety of physiological processes. These actions are mediated via adenosine receptors, of which four different subtypes have now been identified: A_1 , A_{2A} , A_{2B} and A_3 (see Fredholm et al., 1994; Tucker and Linden, 1993). All of these belong to the superfamily of seven transmembrane spanning receptors, but the adenosine receptor subtypes couple to different intracellular signaling pathways (see Fredholm et al., 1994; Tucker and Linden, 1993). In ocular tissue, adenosine has been suggested to regulate intraocular pressure (IOP) (Crosson, 1995; Crosson and Gray, 1996), corneal endothelial ion transport (Riley et al., 1996), rod outer segment phagocytosis by retinal pigment epithelial cells (RP) (Gregory, Abrams and Hall, 1994), retinal acetylcholine release (Blazynski, Woods and Mathews, 1992), retinal and choroidal blood flow (Gidday and Park, 1993a, b; Portellos et al., 1995) and the hypoxic induction of vascular endothelial growth factor (VEGF) and its receptors on retinal capillary endothelial cells and pericytes (Takagi et al., 1996a, b).

Previous pharmacological studies have suggested that A_1 , A_{2A} and A_{2B} receptors may be variously

involved in the regulation of different ocular processes, but selective pharmacological tools are not available for all adenosine receptors and little is known about the cellular distribution of adenosine receptor subtypes in ocular tissues. We have therefore used in situ hybridization with subtype specific adenosine receptor probes to study the expression of A_1 , A_{2A} , A_{2B} and A_3 mRNAs in the rat eye.

2. Materials and Methods

Animals and Tissue Preparation

Adult male Sprague-Dawley rats (12 to 14 weeks of age and each weighing 200 to 250 g) were used in the study. The animals were cared for in accordance with the ARVO Statement for the Use of Animals in Ophthalmic and Vision Research. Animals were killed by a single intraperitoneal overdose of sodium pentobarbital, the eyes were enucleated and immediately frozen on dry ice and stored at -80°C until further processing. Non-pigmented animals were chosen to avoid interference of melanin pigment with the detection of photoemulsion grains.

Generation of A_1 and A_3 Oligonucleotide Probes for in situ Hybridization

Table I summarizes the characteristics of the receptor probes used in this study to detect different adenosine receptor mRNAs in the adult rat eye.

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